Ertuğrul Kundak

Software Engineer | Backend and Full Stack Developer (.NET / Java)

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PROFESSIONAL SUMMARY

I am a Software Engineer with over 2.5 years of experience in banking and financial technology. At Şekerbank I work on Java-based core banking applications, service integrations and Oracle/MSSQL database development. Previously I built secure RESTful APIs for mobile and web applications with ASP.NET Core and Entity Framework; I improved portfolio page performance by 40%, designed a financial analysis dashboard with 30 charts and built an AI-assisted fund recommendation system with the KNN algorithm. On my own time I'm developing BkCompany, a live end-to-end e-commerce platform built with Next.js, TypeScript and PostgreSQL, including Paynet payment, Aras Kargo and e-invoice integrations.

Strengths: system integration, SQL query optimization, JWT-based authentication, performance tuning and data visualization.

TECHNICAL SKILLS

PROGRAMMING LANGUAGES	**C#** · **Java** · **JavaScript** · **SQL** · **T-SQL** · **PL/SQL** · **HTML5** · **CSS3**
BACKEND	**ASP.NET Core** · **.NET Framework** · **ASP.NET MVC** · **Entity Framework** · **LINQ** · **Java (Core Banking)** · **RESTful API** · **Web Services** · **JWT Authentication** · **Middleware** · **N-Tier Architecture**
DATABASE	**Oracle** · **Microsoft SQL Server** · **PostgreSQL** · **Prisma ORM** · **Stored Procedure** · **View** · **Index Management** · **Query Optimization** · **Database Design**
INTEGRATION	**System Integrations** · **Third-Party Services** · **JSON** · **XML** · **API Documentation** · **Payment Integration (3D Secure)** · **Aras Kargo (SOAP) Integration** · **e-Invoice Integration** · **SOAP** · **Cron / Scheduled Jobs** · **Postman**
FRONTEND	**React** · **Next.js** · **TypeScript** · **Tailwind CSS** · **JavaScript (ES6)** · **Responsive Design** · **Highcharts** · **Data Visualization**
AI AND ALGORITHMS	**Machine Learning** · **KNN Algorithm** · **Recommender Systems** · **Data Analysis**
TOOLS AND PROCESSES	**Git** · **GitHub** · **Docker** · **Coolify** · **Jest** · **Visual Studio** · **IntelliJ IDEA** · **VS Code** · **SQL Developer** · **Code Review** · **Error Handling** · **Queue Management**

WORK EXPERIENCE

Software Engineer

Şekerbank T.A.Ş. - Istanbul | December 2025 - Present

- **Core Banking Development:** I build new features and maintain existing modules in Java-based core banking applications.
- **System Integrations:** I develop service/API integrations between in-bank modules and third-party services and manage JSON- and XML-based data exchange.
- **Database Development:** I develop queries, stored procedures and reports on Oracle and Microsoft SQL Server and optimize query performance.
- **Defect Analysis and Support:** I analyze production error logs and take part in root-cause analysis and permanent-fix efforts.
- **Code Quality:** I take part in version control and code review processes, helping maintain coding standards in banking applications.

Software Engineer

İYİGELİR - Istanbul | May 2024 - March 2025

- **Mobile and Web API Development:** I played an active role in RESTful API design, coding and frontend development for mobile and web applications.
- **Performance Optimization:** I improved portfolio page load speed by 40%, noticeably improving the user experience.
- **AI Integration:** I built an AI-assisted fund recommendation system using the KNN algorithm.
- **Data Visualization and Analytics:** I built fund price charts and an analysis dashboard with 30 charts using Highcharts (1–3 second load time).
- **Database and Security:** I optimized query performance by adding indexes to database tables and integrated JWT-based authentication and API security measures.
- **System Architecture:** I built .NET Core middleware solutions, a queue management system and dynamic page structures supporting multiple currencies (USD-EUR-TRY).
- **Error Handling and Quality:** I analyzed and fixed errors reported by users, significantly reducing the overall error rate.

Software Intern

İYİGELİR - Istanbul | February 2024 - May 2024

- **.NET Core Development:** I took development tasks on projects built with the MVC architecture.

- **API Development:** I contributed to updating RESTful APIs and making them modular.
- **Code Optimization:** I carried out code analysis, bug fixing and performance improvement work.
- **Version Control:** I took part in version control processes with Git and GitHub.

Software Intern

PENDC - Bursa | July 2023 - August 2023

- **Frontend Development:** I built a blog site using HTML, CSS and JavaScript.
- **UI Development:** I created the page structure, styling and user interface functions.

PERSONAL PROJECTS

BkCompany — E-Commerce Platform

Next.js 16 · TypeScript · Prisma · PostgreSQL | November 2025 - Present | bkcompany.com.tr

- **Payment Infrastructure:** I built the Paynet 3D Secure payment, installment, saved-card and full-refund flows; stock deduction, order and payment records are written atomically in a single database transaction, and the same transaction reference can never create two payments.
- **Order Lifecycle:** A single-source order state machine, cancellation and refunds, return/exchange requests, withdrawal period counted from delivery, and secure linking of guest orders to an account later.
- **Aras Kargo Integration:** Shipment registration through the Aras SOAP service (SetOrder), box barcode and Code128 dispatch slip generation; reading tracking number, handoff and delivery info back via cron (every 30 min, QT1/QT12) and advancing the order automatically through the state machine. A fallback query strategy for differences between Aras's live responses and its documentation, Turkey-time date conversion, detection of lost shipments with admin notification, closing the Aras record before refunding on cancellation (an ordering that stops a refunded parcel from being shipped), and consignee-pays branch returns.
- **Security and Compliance:** Google OAuth account-linking rules, signed order links, IP-based rate limiting, and recording consent to the distance-sales contract / pre-information form.
- **Other Integrations:** BiFatura e-invoice, Trendyol product/category sync, transactional emails via Resend and AWS SES, abandoned-cart reminders.
- **Deployment and Operations:** Live deployment on Docker and Coolify; scheduled jobs (shipment tracking, cart reminders, notification cleanup), privacy notice and cookie consent compliant with Turkish data-protection law (KVKK), signed email links.
- **Admin Panel and Analytics:** Product/variant, campaign, coupon, collection and customer management; reporting with CLV, cohort and inventory analysis charts.
- **Quality:** 60+ Jest test files, a 40+ table Prisma schema and 35+ migrations; an end-to-end verification flow using fake Paynet/Aras services.

EDUCATION

Software Engineering, Bachelor's

Manisa Celal Bayar University | 2020 - 2024

ACHIEVEMENTS

- **Hackathon - 2nd Place:** My team took 2nd place among 32 teams; I gained experience in highly disciplined teamwork and fast problem solving under limited time.

LANGUAGES

Turkish	English
Native	B1 — Intermediate